Tính các tích phân suy rộng sau:

169. $\int_1^{+\infty} xe^{-x^2}dx$

170. $\int_{\sqrt{2}}^{+\infty} \frac{dx}{x\sqrt{x^2-1}}$

171. $\int_{-\infty}^{+\infty} \frac{dx}{x^2+4x+9}$

172. $\int_{-1}^{+\infty} \frac{dx}{x^2+2x+5}$

173. $\int_e^{+\infty} \frac{dx}{x\ln^3 x}$

174. $\int_{-\infty}^{+\infty} \frac{dx}{x^2+6x+11}$

175. $\int_0^{+\infty} e^{-2x}\cos x dx$

176. $\int_1^e \frac{dx}{x\sqrt{\ln x}}$

177. $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$

178. $\int_0^1 \frac{dx}{x^2+x^4}$

179. $\int_1^e \frac{dx}{x\ln^3 x}$

180. $\int_{1/3}^{2/3} \frac{dx}{x\sqrt{9x^2-1}}$

181. $\int_{-1}^0 \frac{e^{1/x}}{x^3}dx$

182. $\int_{-1}^1 \frac{x-1}{\sqrt[3]{x^5}}dx$

Xét sự hội tụ của các tích phân suy rộng sau:

183. $\int_0^{+\infty} \frac{xdx}{\sqrt{x^5+1}}$

184. $\int_1^{+\infty} \left(1-\cos\frac{2}{x}\right)dx$

185. $\int_1^{+\infty} \frac{1+x^2}{x^3}dx$

186. $\int_{-1}^{+\infty} \frac{xdx}{x^2+\sqrt[3]{x^4+1}}$

187. $\int_2^{+\infty} \frac{dx}{\sqrt{x(x-1)(x-2)}}$

188. $\int_0^{+\infty} \frac{\sin 2x}{x\sqrt{x}+7}$

189. $\int_1^{+\infty} \frac{e^{-x^2}}{x^2}dx$

190. $\int_0^{+\infty} \frac{\sin^2 x}{x}dx$

191. $\int_0^{+\infty} \frac{arctgxdx}{x^3+\sqrt[3]{x^4}}$

192. $\int_0^{+\infty} \frac{xdx}{x^2+4}$

193. $\int_1^{+\infty} \frac{3x^2+\sqrt{(x+1)^3}}{2x^3+\sqrt[3]{x^{10}}}dx$

194. $\int_1^{+\infty} \frac{\cos x+2\sin x}{x}dx$

195. $\int_1^{+\infty} \frac{\sin\frac{1}{x}}{2+x\sqrt{x}}dx$

196. $\int_1^{+\infty} \frac{dx}{\sqrt{x}+\cos^2 x}$